

Heat transfer

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How are things going?

- Schedule
- Website
- Book
- Content
- Exercises
- Solutions

Remember you can find some helpful background info in the appendices:

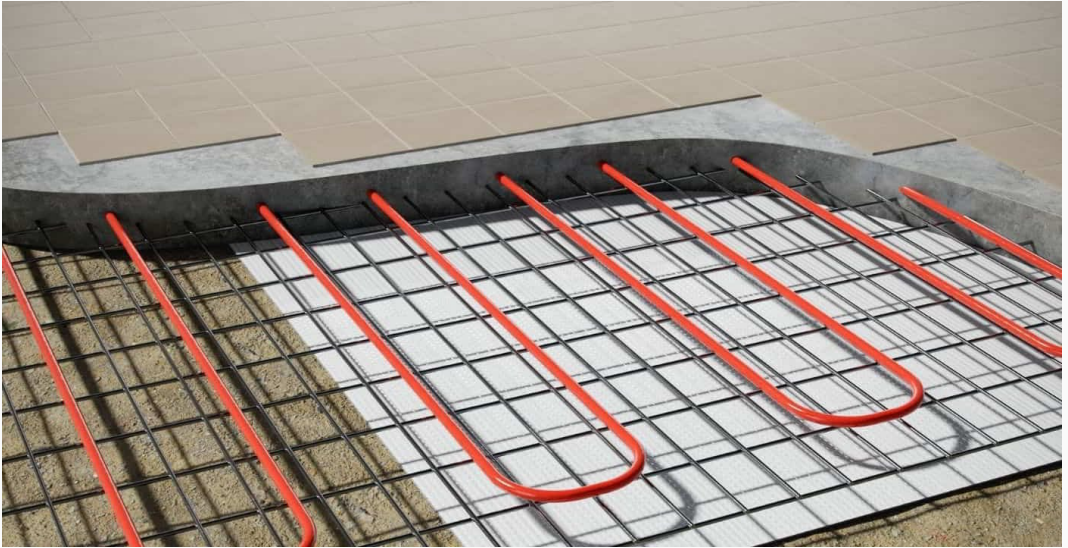
- B. Python data structures
- D. Introduction to VS Code

Be sure to read the assigned text book content *before* class!

Modes

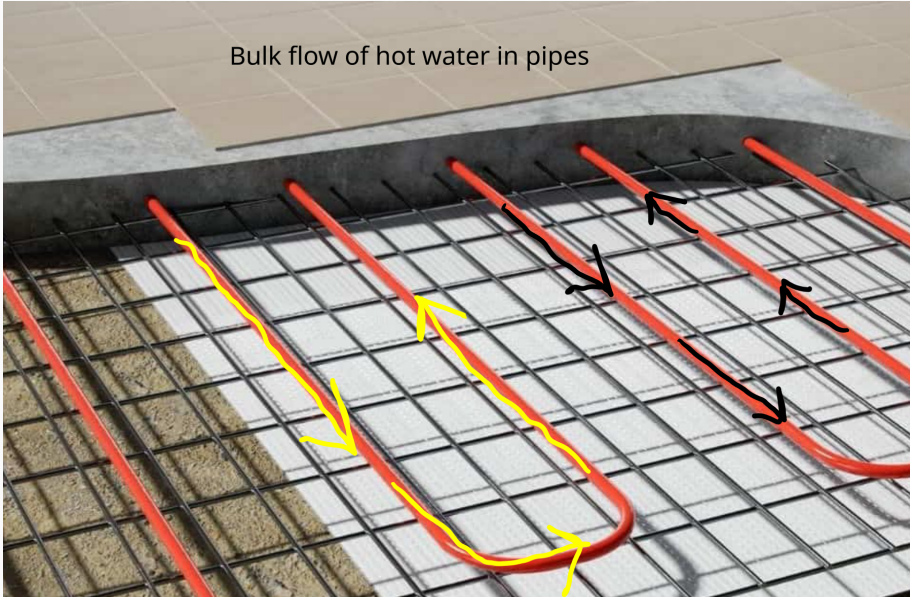
- Advection
- Conduction
- Convection

Advection



Advection

Bulk flow of hot water in pipes



Advection

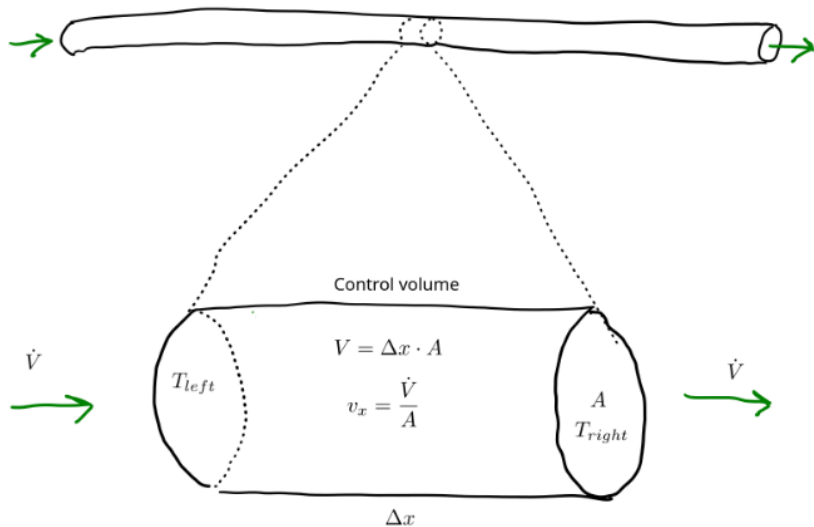


Figure 4.2: A fluid flowing through a constant-diameter round pipe at a constant flow rate.

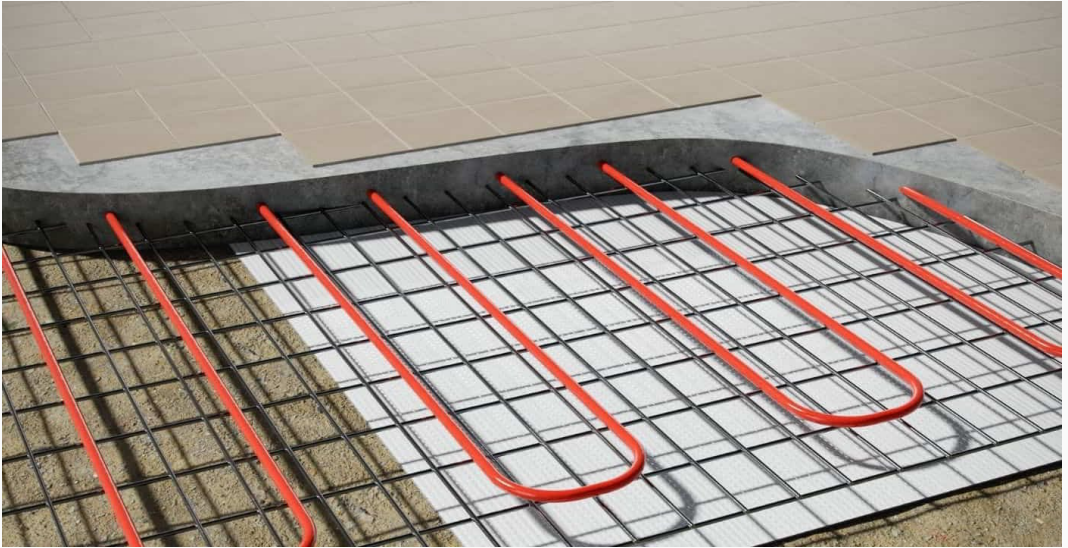
$$\frac{dT}{dt} = -v_x \cdot \frac{(T_{x=\Delta x} - T_{x=0})}{\Delta x}. \quad (4.9)$$

That quotient on the RHS of [Equation 4.9](#) becomes the definition of a derivative as Δx approaches 0, so the associated ODE is:

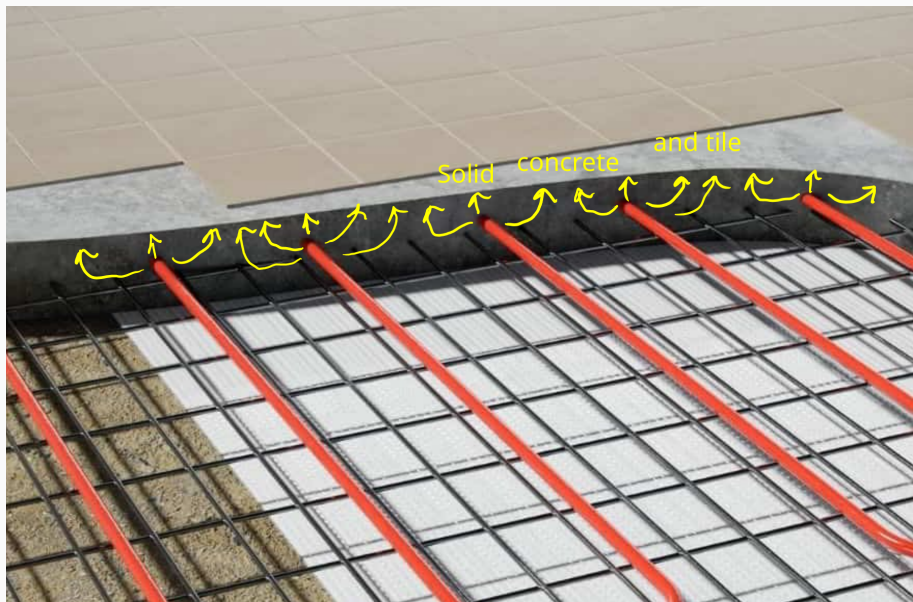
$$\frac{dT}{dt} = -v_x \cdot \frac{dT}{dx}. \quad (4.10)$$

[Equation 4.10](#) is actually pretty intuitive. It says that the rate of temperature change at a fixed location is equal to the temperature derivative over space and the rate at which fluid flows over that same space.

Conduction



Conduction



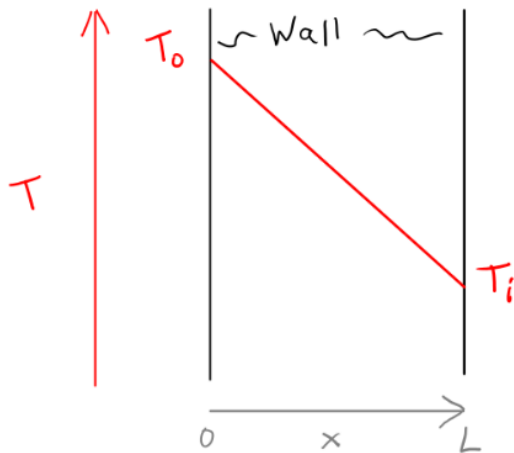


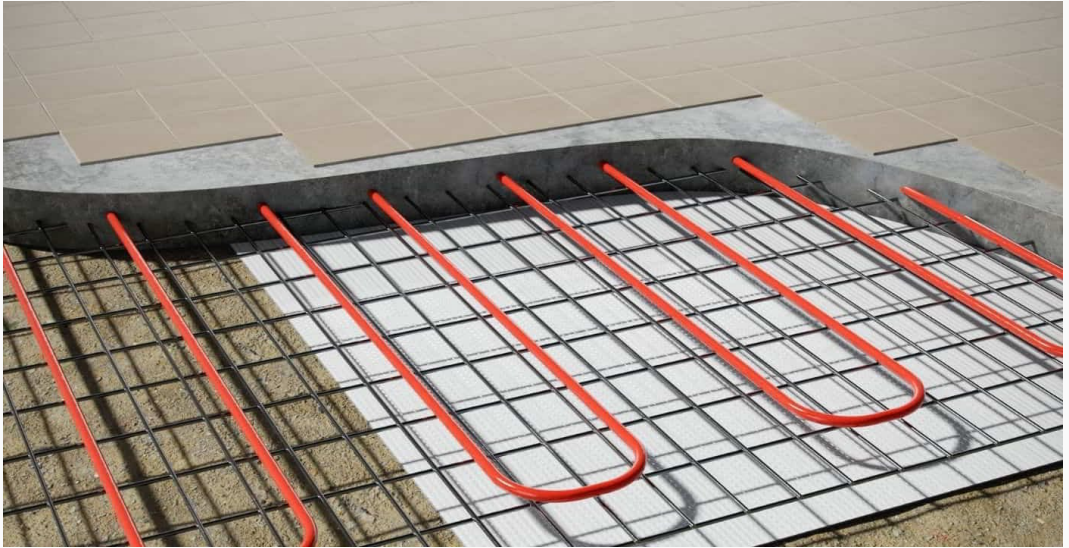
Figure 4.3: A temperature profile through a plane wall

With a linear profile, as in [Figure 4.3](#), Fourier's law can be expressed as

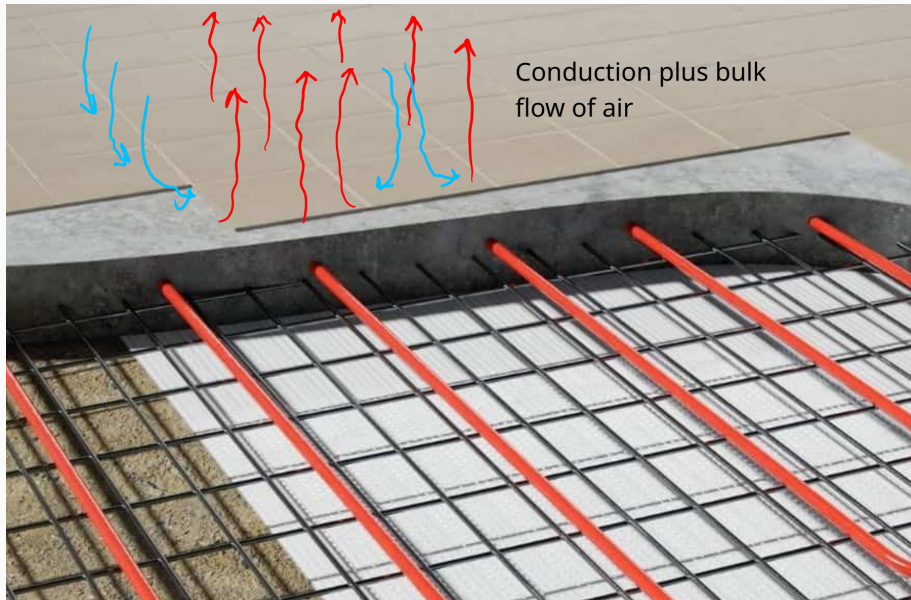
$$q = -k \cdot \frac{\Delta T}{L}, \quad (4.14)$$

where ΔT is the temperature difference across the wall ($T_o - T_i$) and L is the wall thickness. This works because the gradient or derivative is the same everywhere, so can be calculated from the overall difference ΔT and L . For our work in this book, this is the fundamental form of Fourier's law that we will apply.

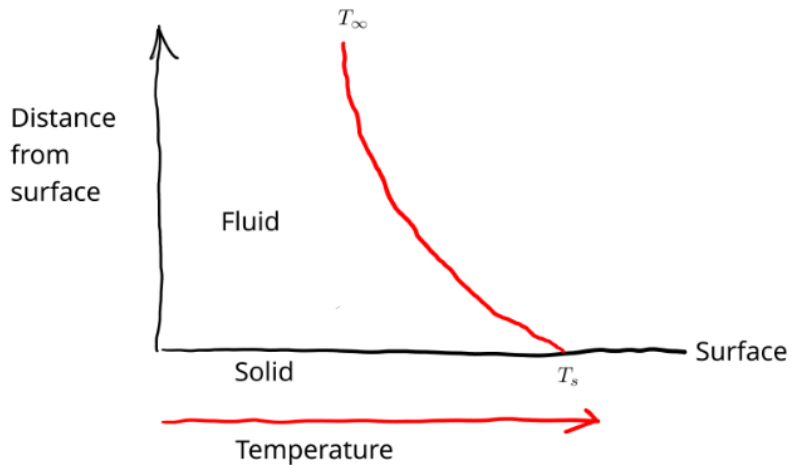
Convection



Convection



Convection



The constitutive equation for convective heat transfer is called Newton's law of cooling:

$$q = h \cdot (T_s - T_\infty). \quad (4.16)$$

- ✓ It is a simple, phenomenological, relationship that posits that the rate of heat loss is proportional to the temperature *difference* (note that distance is not involved) between the surface temperature T_s and fluid temperature T_∞ and some proportionality constant h called the *convection heat transfer coefficient*. We'll often call h a shorter name: convection coefficient. It depends on the physical properties of the fluid as well as whether and how it is flowing—the speed and turbulence. We'll use tabulated values or guesses in our work.